

Warp Drive, When?

Breakthrough Technologies

What are THE 3 breakthroughs we'd like to achieve?

To enable practical interstellar travel, here are **THE 3** breakthroughs that we'll need. These are the goals of the new NASA Breakthrough Propulsion Physics program:

1. Discover new propulsion methods that eliminate or dramatically reduce the need for propellant. This implies discovering fundamentally new ways to create motion, presumably by manipulating gravity or inertia or by manipulating any other interactions between matter and spacetime.
2. Discover how to attain the ultimate achievable transit speeds to dramatically reduce deep space travel times. This implies discovering a means to move a vehicle at or near the actual maximum speed limit for motion through space or through the modification of spacetime itself.
3. Discover fundamentally new on-board energy production methods to power propulsion devices. This third goal is included in the program since the first two goals could require breakthroughs in energy generation to power them and since the physics underlying the propulsion goals is closely linked to energy physics.

What is a breakthrough?

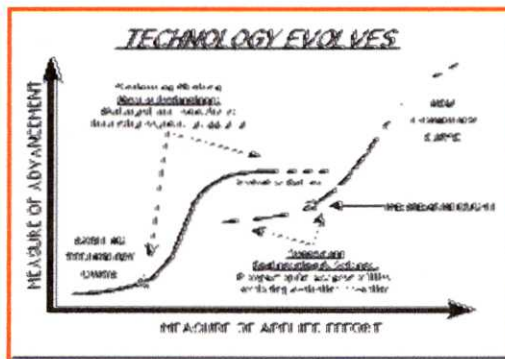


Figure adapted from *Innovation, the Attacker's Advantage*, R. Foster, 1986.

A breakthrough is when the performance limits of an existing device or method are exceeded by a new, different device or method. The key word: **different**.

As technology evolves, a given device or method will reach a point when it can no longer be improved. At this point it has reached the limits of its underlying physical principles. To

exceed this performance limit, a totally different device or method with different underlying physical principles is required. Examples:

- The limits of sailing ships were exceeded with steam ships.
- The speed limits of propeller aircraft were exceeded by jet aircraft
- The altitude limits of aircraft were exceeded by rockets
- The travel limits of rockets will be exceeded by... (to be determined)

The S-Curve figure illustrates both the evolution of a given technology, and the breakthrough event when a new, superior technology becomes viable. For a given technology, the evolution is as follows: Initial efforts result in little advancement and then the technology becomes successful. This success point, at the lower knee of the curve, is where the technology has finally demonstrated its utility. After this point significant progress and improvements are made as several embodiments are produced and the technology becomes widely established. Eventually, however, the physical limits of the technology are

reached, and continued effort results in little additional advancement. This evolution (effort expended versus performance gains) takes the form of an S-Curve. To *go beyond* the limits of the top of a predecessor's S-Curve, a new alternative must be created. This new alternative will have its own S-curve and will eventually require yet another new approach to surpass its performance limits. The **breakthrough event**, is when the new method demonstrates its viability to exceed past the limits of its predecessor.

Paradoxically, it is at the point of diminishing returns of an existing technology when it is most difficult to consider alternatives. Institutions that grow up with a technology become too established, too uniquely adept at their technology to consider alternatives. Alternatives are outside their area of expertise. Established institutions prefer to modify, augment or find new applications for their technology rather than to search for ways to go beyond their technology. Historical evidence shows that this refinement approach does not guarantee sustained market superiority (*Innovation, the Attacker's Advantage*, R. Foster, 1986). If an existing organization wants to avoid its own obsolescence, it must be willing to explore alternatives.

Steam ships were not created by mastering the technologies of sails and riggings. Jet aircraft did not result from mastering piston-propeller aircraft. Transistors were not invented by mastering vacuum tubes. Photocopiers did not result from mastering carbon paper. And breakthrough space drives will not be created by mastering rocket engines.

The work style of pioneering for alternatives is different than the style for building mastery. The main emphasis of day-to-day engineering is to be a master of your chosen technology. Mastery is achieved through continuous improvements; refining, augmenting and finding new applications while sustaining expertise throughout this process. The work style depends on established knowledge and tends to be systematic, relatively predictable, and has a relatively short-term return on investment. Creating new and superior technologies, however, is a wholly different type of work. Going beyond the limits of an existing technology requires a pioneering spirit. It requires imagination to envision future possibilities. Pioneering requires confronting ignorance and creating new knowledge rather than just apply existing knowledge. It requires intuition and subjective judgments to navigate in the absence of an established knowledge base. And because progress is unpredictable and the returns on investment are long-term, it requires the ability to take risks.

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Due to the large number of correspondences that Marc Millis now receives, he regrets that he might not be able to answer more detailed inquiries at this time.

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